

Aristotelis Papatheodorou

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EDUCATION

10/2022 - Ongoing **University of Oxford**, Oxford, United Kingdom

- Ph.D. in Engineering Science, Machine Learning and Articulated Robots Group (ARG), Oxford Robotics Institute (ORI)
 - Funded by University of Oxford's Clarendon Fund Award Scholarship with the Scatchered Scholarship
 - Supported by the J.P. Morgan Chase AI Research Ph.D. Fellowship
 - *Structure Preserving Methods for Physical Intelligence*
- **Activities & Societies:**
 - Member of the Oxford Clarendon Association, Oxford Union Society and St Catherine's College
- **Peer Reviewer:**
 - RSS, IEEE ICRA, IEEE/RSJ IROS, IEEE RA-L, IEEE RA-M, AAAI

01/2016 - 05/2021 **National Technical University of Athens (NTUA)**, Athens, Greece

- Diploma (Integrated Master) in Mechanical Engineering (5-year degree; 315 ECTS)
 - **GPA:** 9.08 / 10 "Excellent" (Top 3% for 2021)
 - **Diploma Thesis:** Design & implementation of a real-time distributed EtherCAT-based motion control system for a multi-DoF quadruped robot, CSL-EP NTUA (Grade: 10 / 10)
 - **Specialization:** Mechanical Design & Automatic Control (Grade: 9.48 / 10)
 - **Significant Projects:**
 - Preliminary design of a novel hybrid Constantly Variable Transmission (CVT) gearbox (Award)
 - Design & modelling of a hydraulic Coordinate Measuring Machine (CMM)
 - Design & modelling of a high-power wind-turbine gearbox
 - A learning-based sensor fusion model for the prediction of cutting tool wear evolution during high-speed milling

07/2018 **University of Duisburg - Essen**, Duisburg, Germany

- Summer School in Efficient Kinematic & Dynamic Modelling of Complex Robotic Systems (Co-organized by DAAD & NTUA)

PROFESSIONAL EXPERIENCE

04/2023 – 01/2024 **Machine Learning Engineer**, (Stealth), San Francisco, CA, USA - Part-Time (Remote)

- Designed and implemented scalable MLOps pipelines for large-scale generative architectures focused on image and video comprehension.
- Experimented and optimized various generative AI architectures, including LLMs, Variational Auto-Encoders (VAEs) and Diffusion models for video understanding and engagement prediction using task-specific priors to increase their inductive bias
- Developed the company's next generation video / image grading tool

04/2016 – 09/2022 **Research Assistant**, Legged Robots Team CSL-EP, NTUA, Athens, Greece

- Designed from scratch multi-sensory (e.g. IMUs) embedded systems in hard real-time, distributed networks
- Developed embedded control systems with concurrent execution schemes, using hardware accelerators
- Developed real-time (Preemptive Kernel) Linux & ROS(2)-based applications
- Implemented high-power, precision motion control solutions with industry leading servo drives
- Modelled & controlled non-linear systems
- Developed system identification algorithms

01/2021 – 06/2022 **Embedded Systems Engineer**, PNOE Inc. (YC W19), Palo Alto, CA, USA - Part Time (Remote)

- Embedded Systems Engineer in the R&D Department of PNOE Inc.
- Developed firmware for the next-generation PNOE device
- Developed real-time capable software for medical applications, in compliance with FDA standards
- Designed various PCBs & electronics
- Executed code verification & validation tasks (Intrusive, non-Intrusive Trace / Static Analysis)

HONORS & AWARDS

- **J.P. MORGAN AI RESEARCH Ph.D. FELLOWSHIP**
 - Highly selective fellowship supporting doctoral research in artificial intelligence.
- **Oxford Clarendon Fund Award Scholarship with the Scatcherd Scholarship**
 - Joint scholarships for full-time DPhil in Engineering Science studies at the University of Oxford
- **NTUA Thomaïdion Award for achieving the highest single-semester GPA for the academic year 2019–2020**
 - Achieved single-semester GPA: 9.71 / 10 (9th semester)
- **NTUA Innovative Design Competition of 2019**, 2nd place
 - Project: The preliminary design of a novel hybrid Constantly Variable Transmission (CVT) gearbox

SKILLS

- **Software** ROS(2), Matlab / Simulink / Simscape / Polyspace, Mathematica, Git, MuJoCo, DS Solidworks, MSC Adams, Autodesk's Eagle, Ansys Mech., PyTorch, Docker, EtherLab / EtherCAT, nRF-Connect
- **Programming Lang.** C / C++, Python, R, Fortran

PUBLICATIONS

Preprints

- **A. Papatheodorou***, P. Vaidhyanathan*, I. Havoutis, N. Ares, and G. J. Milburn, "CALiSYM: Learning Symplectic Dynamics of Real-World Systems through Structured Canonical Lifts," Preprint, 2026 (*Under Review*)
- **A. Papatheodorou**, J. Rojas, I. Havoutis, and C. Mastalli, "Elastic ODYN: Differentiable optimization for infeasible control and learning in robotics," Preprint, 2026 (*Under Review*)
- J. Rojas, **A. Papatheodorou**, S. Martinez, A. Patrizi, I. Havoutis, and C. Mastalli, "ODYN: An all-shifted non-interior-point method for quadratic programming in robotics and AI," Preprint, 2026 (*Under Review*)
- **A. Papatheodorou***, P. Vaidhyanathan*, D. R. M. Arvidsson-Shukur, M. T. Mitchison, N. Ares, and I. Havoutis, "QuantGraph: A receding-horizon quantum graph solver," Preprint, 2025 (*Under Review*)

International Conference Proceedings

- **A. Papatheodorou**, W. Merkt, A. L. Mitchell, and I. Havoutis, "Momentum-aware trajectory optimisation using full-centroidal dynamics and implicit inverse kinematics," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Abu Dhabi, UAE, Oct. 2024.
- A. L. Mitchell, W. Merkt, **A. Papatheodorou**, I. Havoutis, and I. Posner, "Gaitor: Learning a unified representation across gaits for real-world quadruped locomotion," in *Proc. 8th Conf. Robot Learn. (CoRL)*, Proc. Mach. Learn. Res., vol. 270, PMLR, Nov. 2025.
- A. Mastrogeorgiou, **A. Papatheodorou**, K. Koutsoukis, and E. Papadopoulos, "Learning energy-efficient trotting for legged robots," in *Proc. CLAWAR*, Ponta Delgada, Portugal, Sep. 2022.
- **A. Papatheodorou**, A. Mastrogeorgiou, K. Machairas, K. Koutsoukis, and E. Papadopoulos, "Middleware interconnecting ROS/ROS2 with the EtherCAT protocol," in *Proc. ROSCon (ROS World)*, Oct. 2021.

Journal & Book Chapter Publications

- **A. Papatheodorou***, P. Vaidhyanathan*, M. T. Mitchison, N. Ares, and I. Havoutis, "MetaSym: A symplectic meta-learning framework for physical intelligence," *Trans. Mach. Learn. Res.*, 2026.
- A. Mastrogeorgiou, **A. Papatheodorou**, K. Koutsoukis, and E. Papadopoulos, "Learning energy-efficient trotting for legged robots," in *Robotics in Natural Settings*, Lecture Notes in Networks and Systems, vol. 530, Cham, Switzerland: Springer, 2022, pp. 204–215.

Conference Workshops & Posters

- **A. Papatheodorou***, P. Vaidhyanathan*, N. Ares, and I. Havoutis, "Learning physical systems: Symplectification via gauge fixing in Dirac structures," presented at *Equivariant Syst.: Theory Appl. State Estimation, Artif. Intell. Control Workshop, Robot. Sci. Syst. (RSS)*, Los Angeles, CA, USA, Jun. 2025.
- **A. Papatheodorou**, A. Mastrogeorgiou, K. Machairas, K. Koutsoukis, and E. Papadopoulos, "Real-time EtherCAT master for ROS," presented at *ROS World (Lightning Talk)*, Nov. 12, 2020.

Thesis

- **A. Papatheodorou**, "Design & implementation of a real-time distributed EtherCAT-based motion control system for a multi-DoF quadruped robot," Integrated M.S. thesis, National Technical Univ. Athens, Athens, Greece, 2021. [Online]. Available: <https://dspace.lib.ntua.gr/xmlui/handle/123456789/53738>